

Table 2. Robustness and detectability on Llama-2-7B across attack conditions. For each condition, we report AUROC (AUC), TPR at 1% FPR, and the keyless z-score ((Liu et al., 2025)). Higher AUC and TPR indicate greater robustness, whereas lower z-scores indicate lower detectability under this detector. Superscripts identify watermark families: ^B Biased, ^F Bias-free, ^S Semantic, ^D Dist-preserving, ^W Training-time, and * Hybrid. DAWA is shown separately as a co-designed baseline. The summarization column ([†]) reports the results of the summarization attack from *WaterJudge* with $\hat{\epsilon} \approx 0.55$.

	No attack			DIPPER ($\hat{\epsilon} \approx 0.25$)			OPT-2.7B ($\hat{\epsilon} \approx 0.15$)			WM-removal ($\hat{\epsilon} \approx 0.15$)			Synonym ($\hat{\epsilon} \approx 0.15$)			Back-trans. ($\hat{\epsilon} \approx 0.42$)			Summ. [†] ($\hat{\epsilon} \approx 0.55$)		
Method	AUC	TPR	z	AUC	TPR	z	AUC	TPR	z	AUC	TPR	z	AUC	TPR	z	AUC	TPR	z	AUC	TPR	z
KGW ^B	.99	1.00	30.1	.86	.64	9.6	.78	.59	8.4	.78	.59	8.1	.78	.59	8.3	.58	.52	−1.2	.52	.14	−2.3
Unigram ^B	.99	1.00	11.2	.88	.67	8.8	.79	.62	7.9	.79	.61	7.6	.79	.62	7.8	.57	.51	−1.0	.53	.15	−2.0
DiPMark ^F	.99	1.00	43.2	.90	.80	3.9	.91	.86	3.6	.90	.85	3.4	.91	.86	3.5	.60	.54	−0.8	.55	.18	−1.1
HCW ^F	.99	1.00	105	.91	.82	3.4	.92	.88	3.1	.92	.88	3.0	.92	.88	3.0	.59	.53	−0.6	.56	.20	−0.8
HeavyWater ^F	.99	1.00	38.5	.88	.76	4.2	.89	.82	3.8	.88	.81	3.5	.89	.82	3.6	.56	.50	−1.0	.53	.16	−1.3
SimplexWater ^F	.99	1.00	35.2	.87	.74	4.5	.88	.80	4.0	.87	.79	3.7	.88	.80	3.8	.55	.49	−1.1	.52	.15	−1.4
SemStamp ^S	.98	.98	8.5	.93	.87	2.8	.94	.91	2.4	.93	.90	2.2	.94	.91	2.3	.82	.75	1.5	.66	.38	0.5
PMark ^S	.99	.99	7.2	.94	.89	2.5	.95	.93	2.1	.94	.92	1.9	.95	.93	2.0	.85	.79	1.2	.64	.43	0.3
SimMark ^S	.98	.97	9.1	.92	.85	3.0	.93	.89	2.6	.92	.88	2.4	.93	.89	2.5	.80	.72	1.7	.64	.35	0.7
CGW ^D	.99	1.00	−5.8	.50	.15	−10	.50	.31	−5.4	.50	.31	−5.6	.50	.31	−5.5	.51	.21	−6.5	.50	.16	−6.9
GaussMark ^W	1.00	1.00	12.4	.82	.71	8.6	.85	.76	7.8	.83	.74	7.3	.84	.75	7.5	.62	.55	2.1	.56	.21	0.4
DAWA	1.00	1.00	2.2	.54	.03	1.6	.74	.05	1.3	.71	.07	1.4	.84	.08	3.0	.75	.03	2.0	.63	.02	1.0
Hybrid*	.99	1.00	−7.8	.94	.89	4.2	.96	.93	3.8	.95	.92	4.3	.95	.93	4.4	.86	.79	1.8	.61	.47	0.6